Ji-Hoon Oh

Affiliation: Scripps Institution of Oceanography, UC San Diego, San Diego, CA, United States
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Professional Experience

Dec 2024 – Present

Postdoctoral Researcher @ Scripps Institution of Oceanography, UC San Diego

- Location: San Diego, United States
- Supervisor: Prof. Shang-Ping Xie

Mar 2024 – Nov 2024

Postdoctoral Researcher @ School of Earth & Environmental Sciences, Seoul National University

- Location: Seoul, Republic of Korea
- Supervisor: Prof. Jong-Seong Kug

Education

Mar 2021 – Feb 2024

Ph.D. @ Department of Environmental Science and Engineering, Pohang University of Science and Technology (POSTECH)

- Location: Pohang, Republic of Korea
- Dissertation: Irreversibility of global climate system and its potential predictability
- Advisor: Prof. Jong-Seong Kug

Mar 2019 – Feb 2021

M. S. @ Department of Environmental Science and

Engineering, Pohang University of Science and Technology (POSTECH)

- Location: Pohang, Republic of Korea
- Thesis: Impact of Antarctic meltwater forcing on East Asian climate under global warming
- Advisor: Prof. Jong-Seong Kug

Mar 2013 – Feb 2019 **B. S. @ Atmospheric Science, Pusan National University**

- Location: Pusan, Republic of Korea
- Mandatory Military Service: Republic of Korea Navy (Mar 2014 – Feb 2016)

Grants

- **Grant (RS-2025-02653909) from National Research Foundation of Korea (NRF) Basic Science Research Program**
  : Evaluation of Climate Change Uncertainty Post-Carbon Neutrality: Forthcoming Tipping Point of the Atlantic Ocean Circulation (RS-2025-02653909), Principal Investigator, ~43,000 USD, Dec 2025 – Nov 2026

Research Interests

- Large-scale climate dynamics
- Climate responses under net-zero or negative emissions
- Tipping Point
- Atlantic Meridional Overturning Circulation
- Earth System Modeling

Awards

- **Best Poster Award** from Korea Hydrographic and Oceanographic Agency (KHOA), Jun 2019
- **Best Poster Award** from Korean Meteorological Society (KMS), Apr 2023
- **Sung-Kee Chung Award** from Pohang University of Science and Technology (POSTECH), Feb 2024

  : POSTECH Honor (Most Outstanding Ph.D. Graduate in the Field of Natural Science)
- **Best Dissertation Award** from Korean Meteorological Society (KMS), Oct 2024

Publications

†: equally contributed

* : corresponding author

Published

[2026]

22. Emergence of unprecedented North Atlantic salinity extremes under AMOC slowdown, *Nature Communications* (2026) [in press] **[link]**
T. Iwakiri*, J.-S. Kug*, S.-I. An, Y. Shin, **J.-H. Oh**, S. Wang

21. Critical role of low cloud feedback in irreversible sea level rise, *Nature Communications* (2026) [in press] **[link]**
S. Wang, Y. Shin, **J.-H. Oh**, H. Kim, J.-S. Kug*

20. Biophysical impacts of urbanization on climate change and vegetation in Borneo island, *Journal of Environmental Management* (2026), 398, 128446 **[link]**
D. Lee, **J.-H. Oh**, J. Kam, S.-W. Park, J.-S. Kug*

[2025]

19. Noise-induced tipping of Atlantic Meridional Overturning Circulation under climate mitigation scenarios (2025), *Nature Communications*, 16, 11515 **[link]**
J.-H. Oh, J.-S. Kug*, Y. Shin, X. Geng, S. Wang, F.-F. Jin, S.-I. An, S.-P. Xie, W. Liu

18. Arctic stratospheric ozone as a precursor of ENSO events since 2000s (2025), *npj Climate and Atmospheric Science, 8, 1, 338* **[link]**
J.-H. Park*, J.-H. Koo*, J.-S. Kug*, S.-J. Lee, M.-K. Sung, J. Kim, E.-C. Chang, Y.-M. Yang, S.-S. Park, K.-H. Kwak, **J.-H. Oh**, H.-J. Kang, S.-I. An

17. Nonmonotonic future changes in the North Atlantic warming hole under a fast CO_2 emission scenario (2025), *Journal of Climate, 38, 21, 6011-6023* **[link]**
X. Geng, **J.-H. Oh**, Y. Shin, J. Shin, K.-M. Noh, S.-W. Park, J.-S. Kug*

16. Land aridification persists in vulnerable drylands under climate mitigation scenarios (2025), *Communications Earth & Environment, 6, 1, 732* **[link]**
J. Piao, W. Chen*, J.-S. Kug*, S. Chen, **J.-H. Oh**, L. Wang, Q. Cai

15. Phytoplankton blooming mechanisms over the East China Sea during post-El Niño summers (2025), *Biogeosciences, 22, 13, 3165-3180* **[link]**
D.-G. Lee, **J.-H. Oh**, J. Kam, J.-S. Kug*

14. Emergence of an oceanic CO_2 uptake hole under global warming (2025), *Nature Communications, 16, 1, 3199* **[link]**
H. Lee, K.-M. Noh, **J.-H. Oh**, S.-W. Park, Y. Shin*, J.-S. Kug*

[2024]

13. Pervasive fire danger continued under a negative emission scenario (2024), *Nature Communications, 15, 1, 10010* **[link]**
H.-J. Kim, J.-S. Kim*, S.-I. An*, J. Shin, **J.-H. Oh**, J.-S. Kug

12. Fast recovery of North Atlantic sea level in response to atmospheric CO_2 removal (2024), *Communications Earth & Environment, 5, 1, 652* **[link]**
S. Wang, Y. Shin, **J.-H. Oh***, J.-S. Kug*

11. Delayed ENSO impact on phytoplankton variability over the Western-North Pacific Ocean (2024), *Environmental Research Communications, 6, 10, 101006* **[link]**
D.-G. Lee, **J-H. Oh***, J.-S. Kug*

10. Deep ocean warming-induced El Niño changes (2024), *Nature Communications, 15, 1, 6225* **[link]**
G.-I. Kim, **J.-H. Oh**, N.-Y. Shin, S.-I. An, S.-W. Yeh, J. Shin, J.-S. Kug*

9. Fast and Slow Responses of Atlantic Meridional Overturning Circulation to Antarctic Meltwater Forcing (2024), *Geophysical Research Letters*, 51, 9, e2024GL108272 **[link]**
Y. Shin, X. Geng, **J.-H. Oh**, K.-M. Noh, E. K. Jin, J.-S. Kug*

8. Emergent climate change patterns originating from deep ocean warming in climate mitigation scenarios (2024), *Nature Climate Change, 14, 3, 260-266* **[link]**
J.-H. Oh, J.-S. Kug*, S.-I. An, F.-F. Jin, M. McPhaden, J. Shin

7. Role of Atlantification in enhanced primary productivity in the Barents Sea (2024), *Earth's Future, 12, 1, e2023EF003709* **[link]**
K.-M. Noh, **J.-H. Oh**, H.-G. Lim, J.-S. Kug*

[2023]

6. Increase in convective Extreme El Nino events in the CO_2 removal scenario (2023), *Science Advances, 9, 25, eadh2412* **[link]**
Pathirana. G, **J.-H. Oh**, W. Cai, S.-I. An, S.-K. Min, S.-Y. Jo, J. Shin, J.-S. Kug*

5. What controls the future phytoplankton change over Yellow and East China Seas under global warming? (2023), *Frontiers in Marine Science, 10, 1010341* **[link]**
D.-G. Lee, **J.-H. Oh**, K.-M. Noh, E.-Y. Kwon, Y.-H. Kim, J.-S. Kug*

[2022]

4. Centennial memory of the Arctic Ocean for future Arctic climate recovery in response to carbon dioxide removal (2022), *Earth's Future, 10, 8, e2022EF002804* **[link]**
J.-H. Oh, S.-I. An, J. Shin, J.-S. Kug*

3. Antarctic meltwater-induced dynamical changes in phytoplankton in the Southern Ocean (2022), *Environmental Research Letters, 17, 2, 024022* **[link]**
J.-H. Oh, K.-M Noh, H.-G Lim, E. K. Jin, S.-Y. Jun, J.-S. Kug*

2. Hysteresis of intertropical convergence zone to CO_2 forcing (2022), *Nature Climate Change, 12, 1, 47-53* **[link]**
J.-S. Kug†*, **J.-H. Oh**†, S.-I. An, S.-W. Yeh, S.-K. Min, S.-W. Son, J. Kam, Y.-G. Ham, J. Shin

[2020]

1. Impact of Antarctic meltwater forcing on East Asian climate under greenhouse warming (2020), *Geophysical Research Letters, 47, 21, e2020GL089951* **[link]**
J.-H Oh, W.-S. Park, H.-G. Lim, K.-M. Noh, E. K. Jin, J.-S. Kug*

In Progress

6. Antarctic meltwater amplifies nutrient depletion under global warming (*submitted*)
D.-G. Lee, **J.-H. Oh**, E.-Y. Kwon, J.-S. Kug*

5. Reconciled warning signals in observations and models imply approaching AMOC tipping point (*submitted*)
Y. Shin, **J.-H. Oh**, N. Boers*, S. Bathiany, M. Ben-Yami, M. Årthun, H. Lee, H. Kim, T. Iwakiri, G.-I. Kim, H. Kim, J.-S. Kug*

4. Deep learning highlights underestimated risk of AMOC collapse in current climate models (*under review*)
Y. Shin, N. Boers, Y. Huang, J. Ko, **J.-H. Oh**, J.-S. Kug*

3. Centennial southward migration of the tropical rain belt in a post-zero emissions world (*under review*)
J.-H. Oh* and S.-P. Xie

2. Atlantic Meridional Overturning Circulation slowdown amplifies decadal variability along the Kuroshio-Oyashio and Gulf Stream extensions (*under review*)
J.-H. Oh, S.-P. Xie*, J.-S. Kug, J.-H. Park, Y. Yamagami, A. Miyamoto, G.-I. Kim, B. Wu,

1. Climate irreversibility and transient hysteresis under CO_2 removal (*in revision;* *review paper)
J.-S. Kug*, Y. Shin, **J.-H. Oh**, C. Liu, S.-S. Son, S.-W. Yeh, S.-I. An, S.-K. Min, J. Schwinger, H. Lee, N.J. Steinert, K. Zickfeld